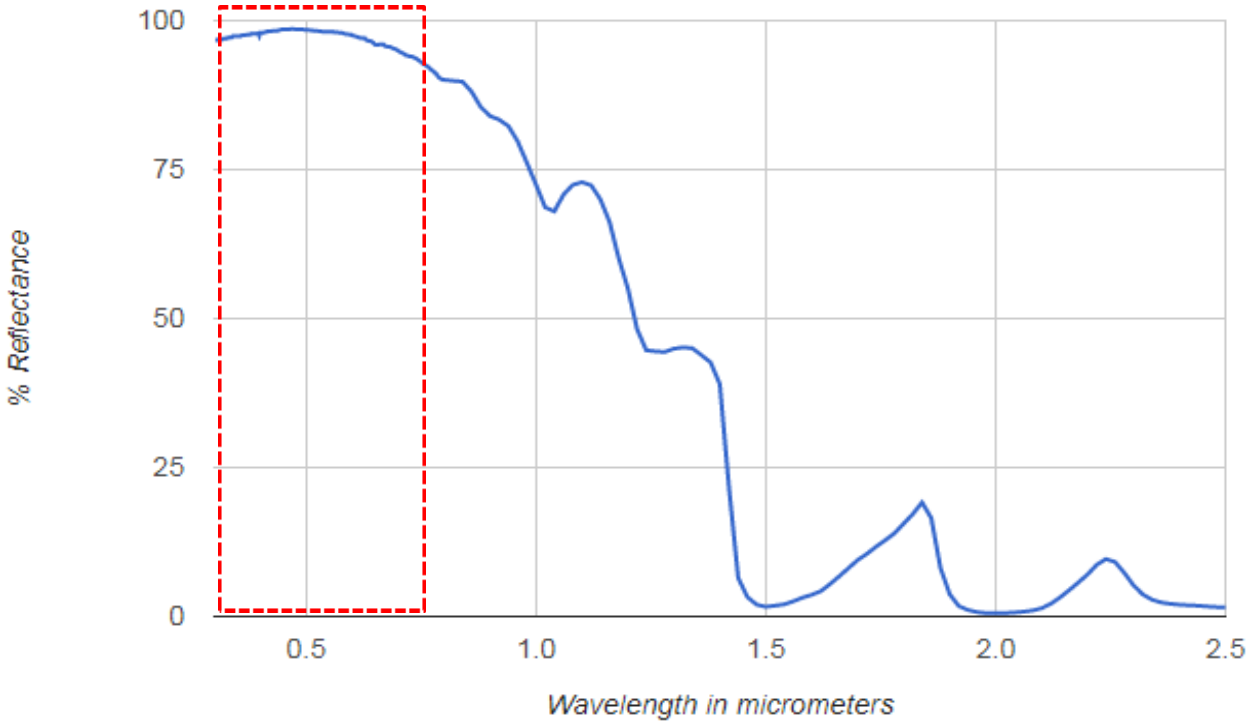

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Spectral Reflectance of Snow



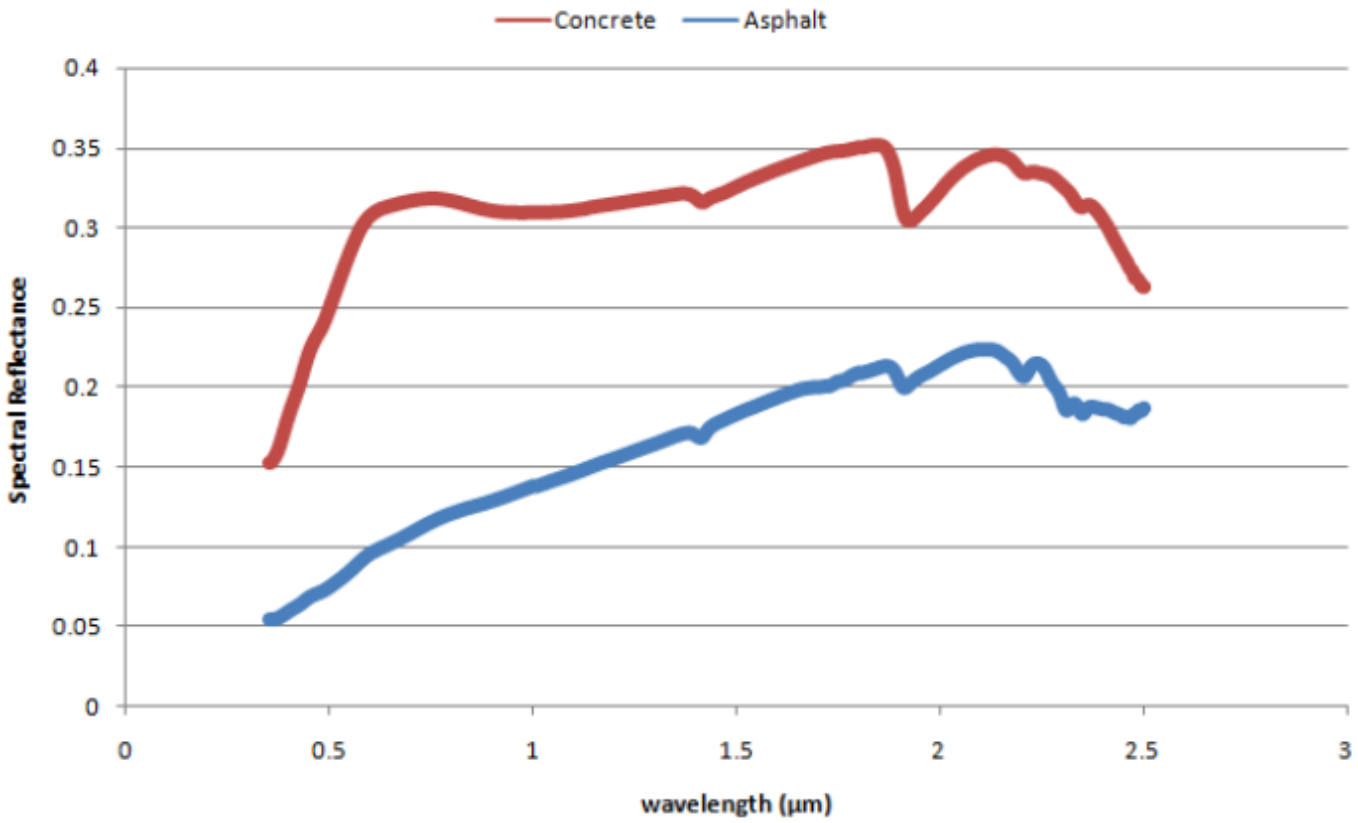
- Wavelength
- Age (time)
- Grain size
- Depth and density of the snowpack
- Size and amount of the impurities

↑ Fresh to Old ↓

Grain Size

↑ Small to Large ↓

Spectral Reflectance- Impervious Surfaces



- Wavelength
- Age
- Type of the material
- Surface Condition

SPECTRAL REFLECTANCE OF SOIL

SOIL CHARACTERISTICS

- **In RS data, soils may be distinguished by their varying spectral responses**
- **Texture and physical properties of soils results in unique spectral response**
- **The factors that influence spectral response includes:**
  - **Particle size (surface roughness)**
  - **Moisture content**
  - **Organic matter**
  - **Soil constituents (proportion of sand, silt, and clay),**
  - **Soil Color**

SPECTRAL REFLECTANCE CHARACTERISTICS

- **These factors are complex, variable, and interrelated.**
- **For example, the presence of moisture in the soil will decrease its reflectance**
- **Soil moisture content is strongly related to the soil texture: coarse, sandy soils are usually well-drained, resulting in low moisture content and relatively high reflectance.**
- **Poorly drained fine-textured soils will generally have lower reflectance**

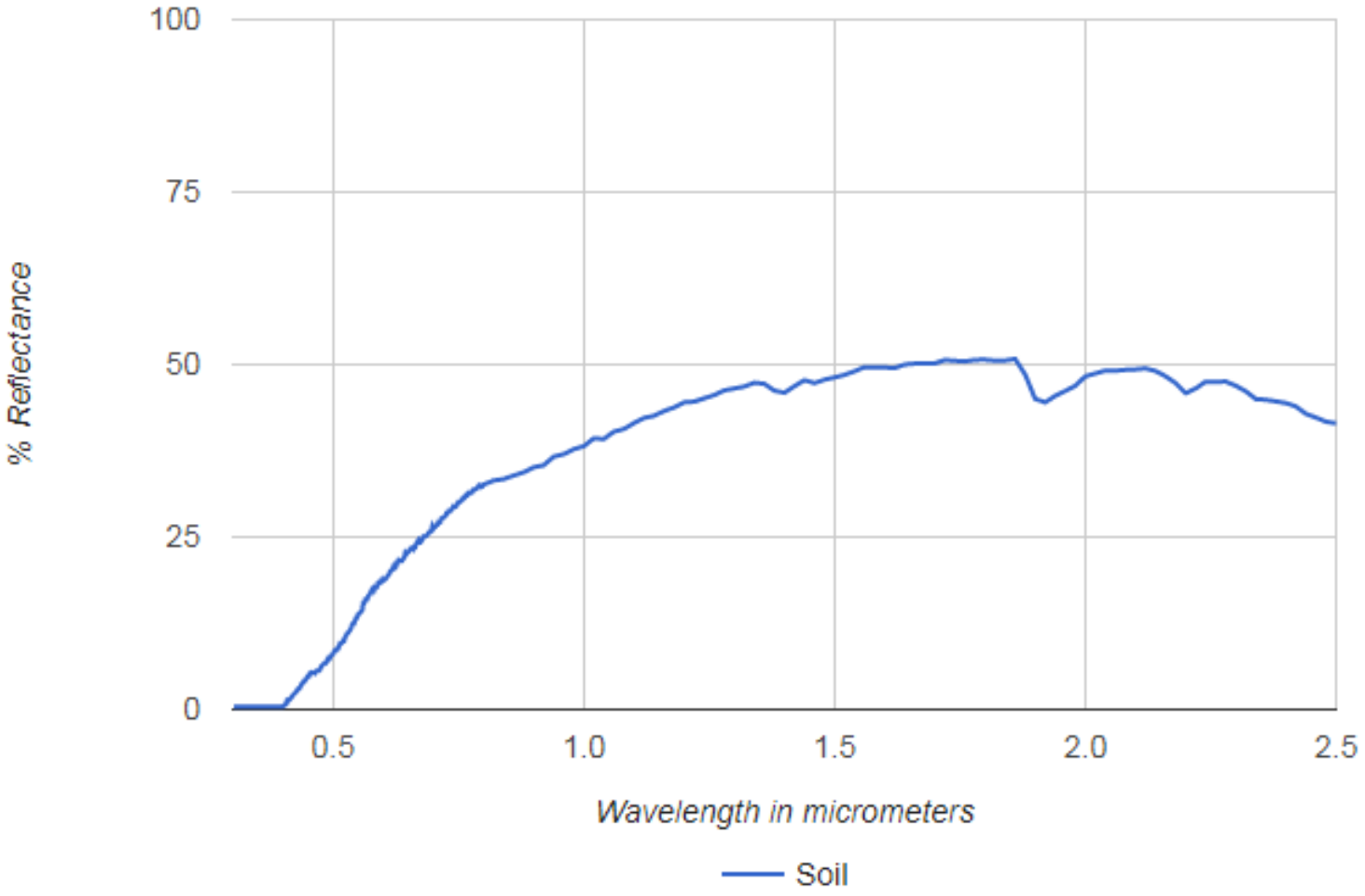
SPECTRAL REFLECTANCE CHARACTERISTICS

- **In the absence of water, the soil exhibits the reverse tendency:**
  - **Coarse-textured soils will appear darker than fine-textured soils. Thus, the reflectance properties of soil are consistent only within particular ranges of conditions**
- **Organic matter reduces soil reflectance**

SOIL CHARACTERISTICS

- **Soils with organic matter content are often dark colored**
- **Moist soils have nearly uniform dark tones and low reflectance**
- **Dry soils give a light tone and high reflectance**
- **Marshy areas can easily be identified due to the high absorption of EMR by water and give dark black tone**
- **Saline areas -high reflectance (white tone)**

General Spectral Reflectance of



Moisture Content

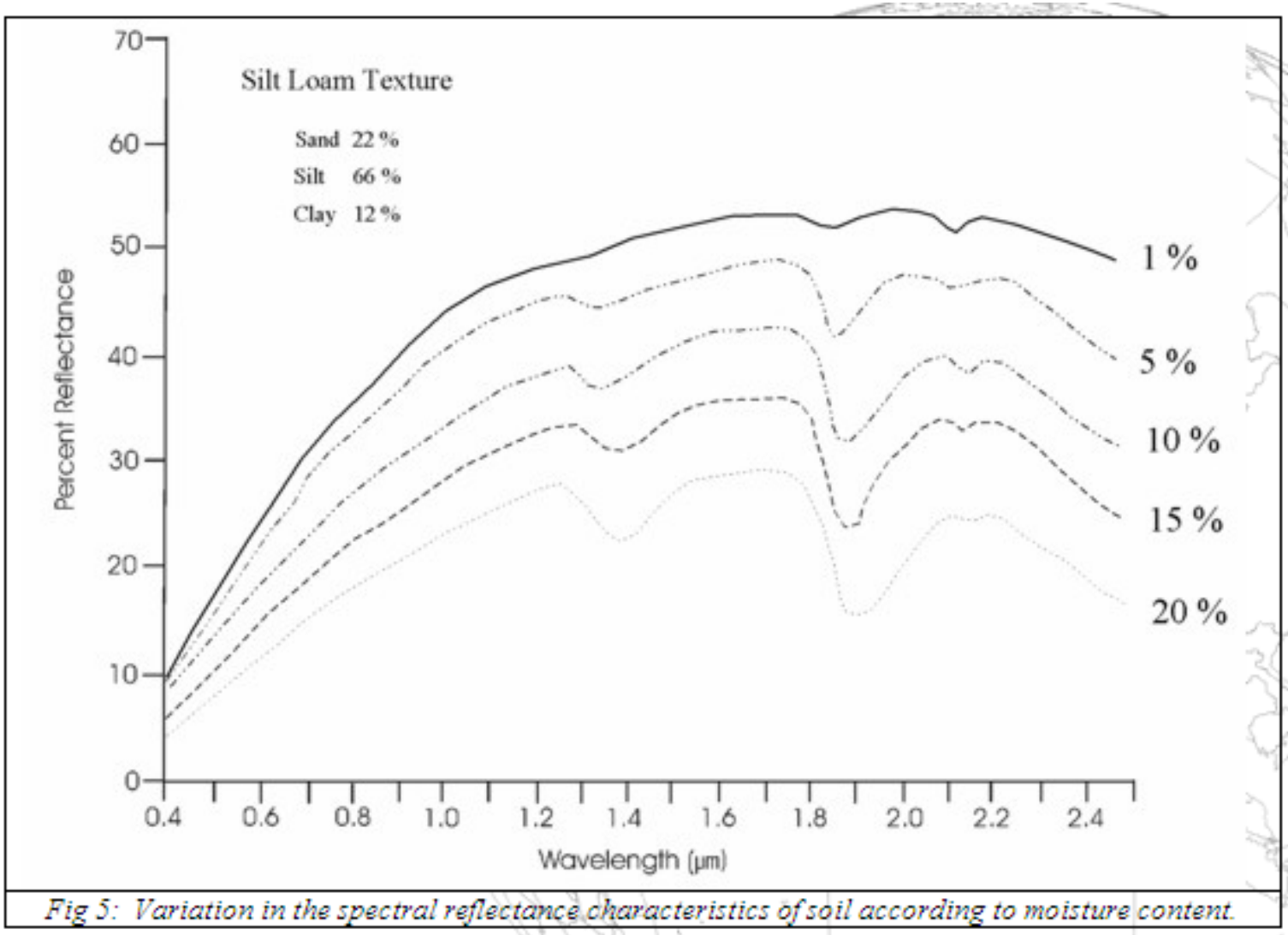


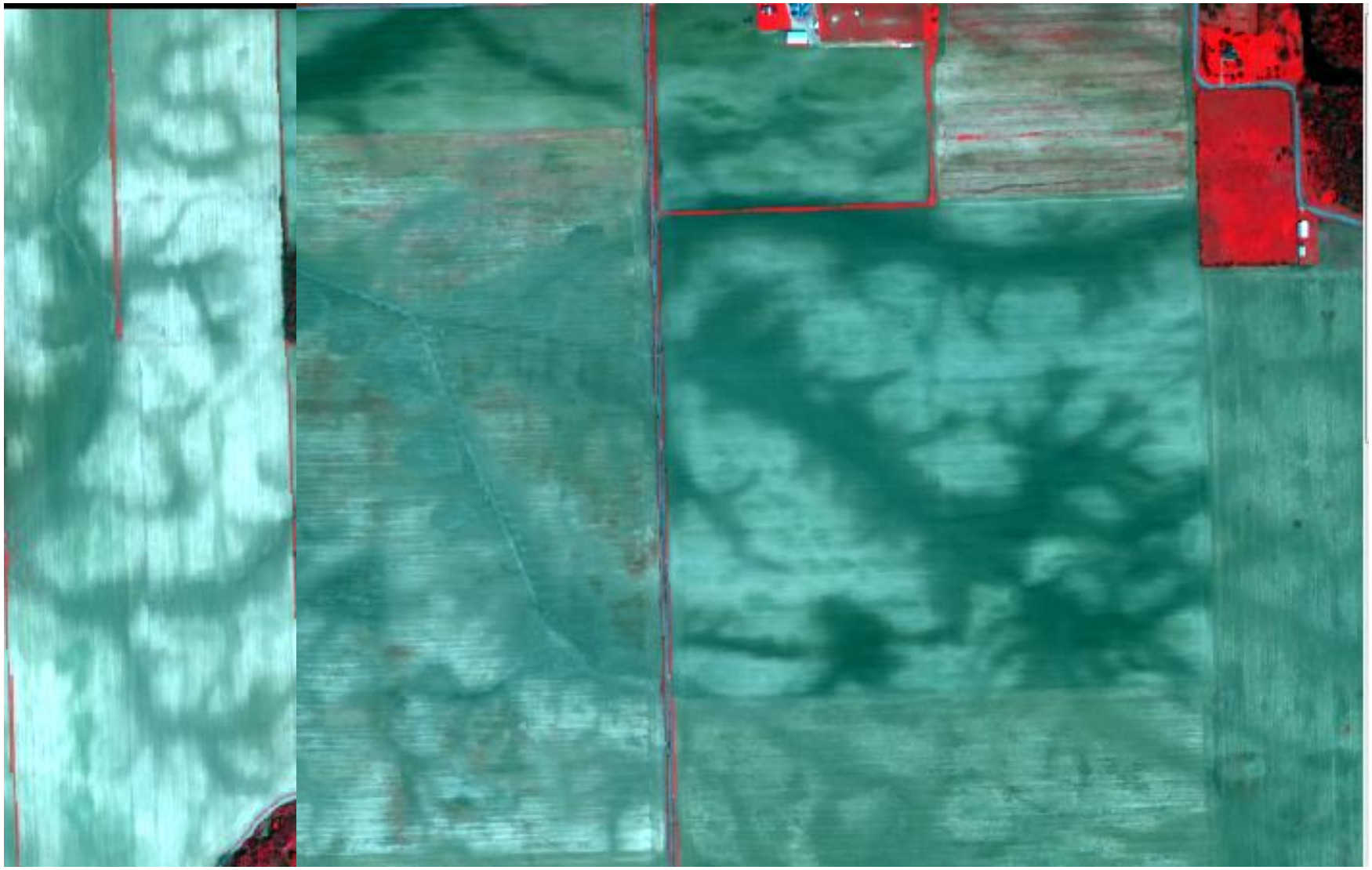
Fig 5: Variation in the spectral reflectance characteristics of soil according to moisture content.

http://www.cps-amu.org/sf/notes/m1r-1-8.htm

Wet/Moist Soils

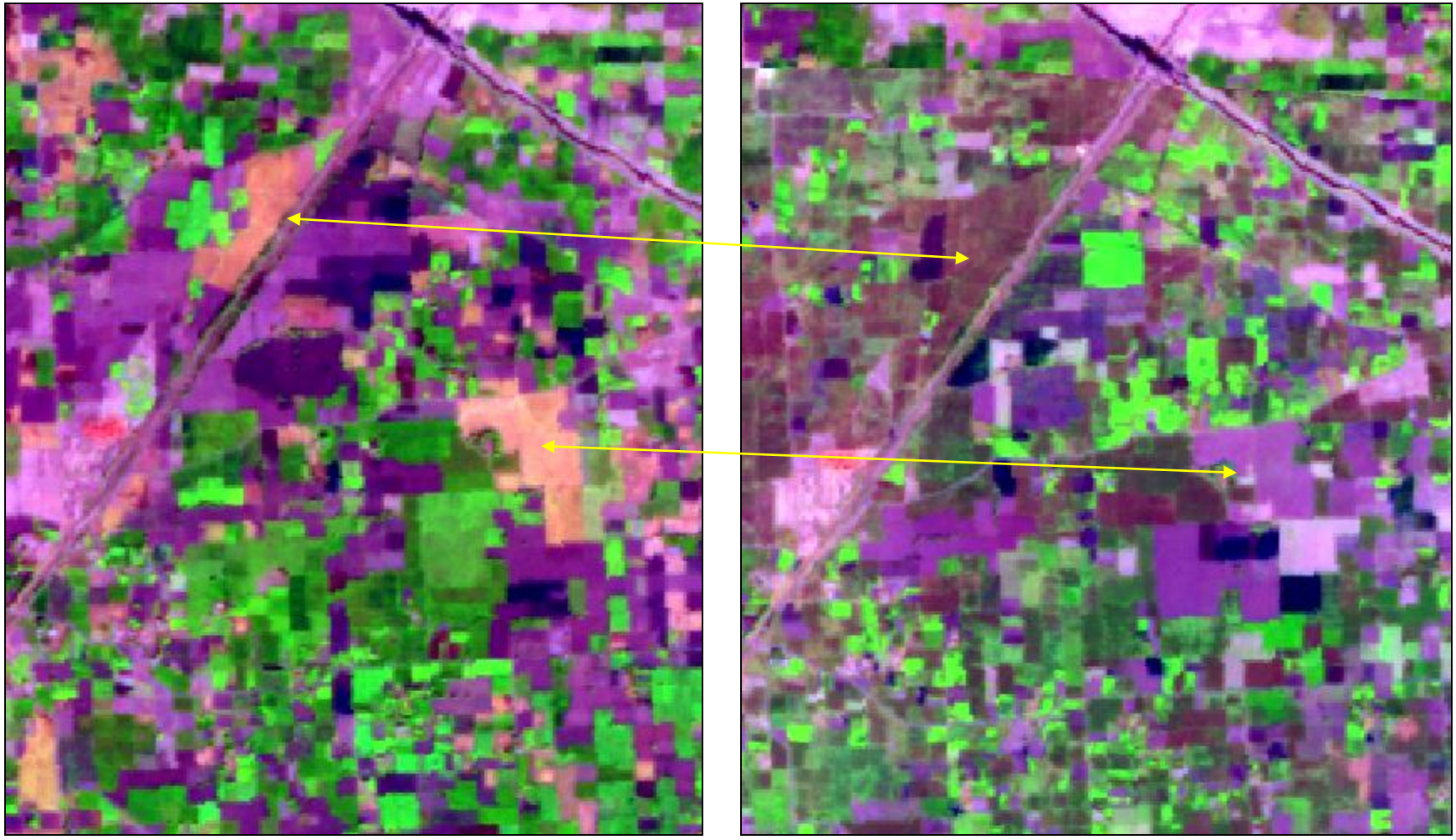


Wet/Moist Soils



Dark Color: Wet/Moist Soils
Bright: Dry Soils

Spectral Reflectance –Temporal Effects



September **April**

Particle Size

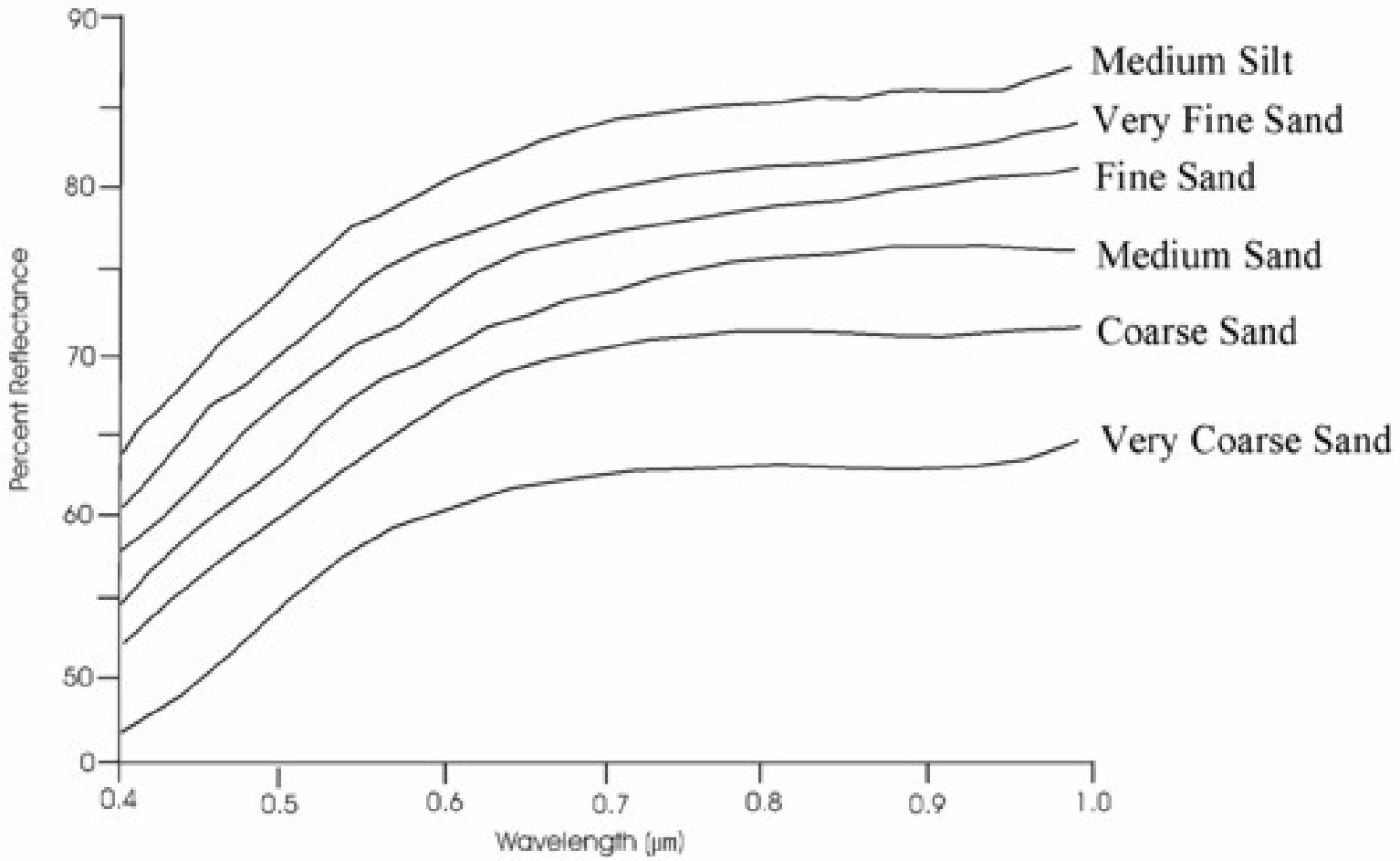
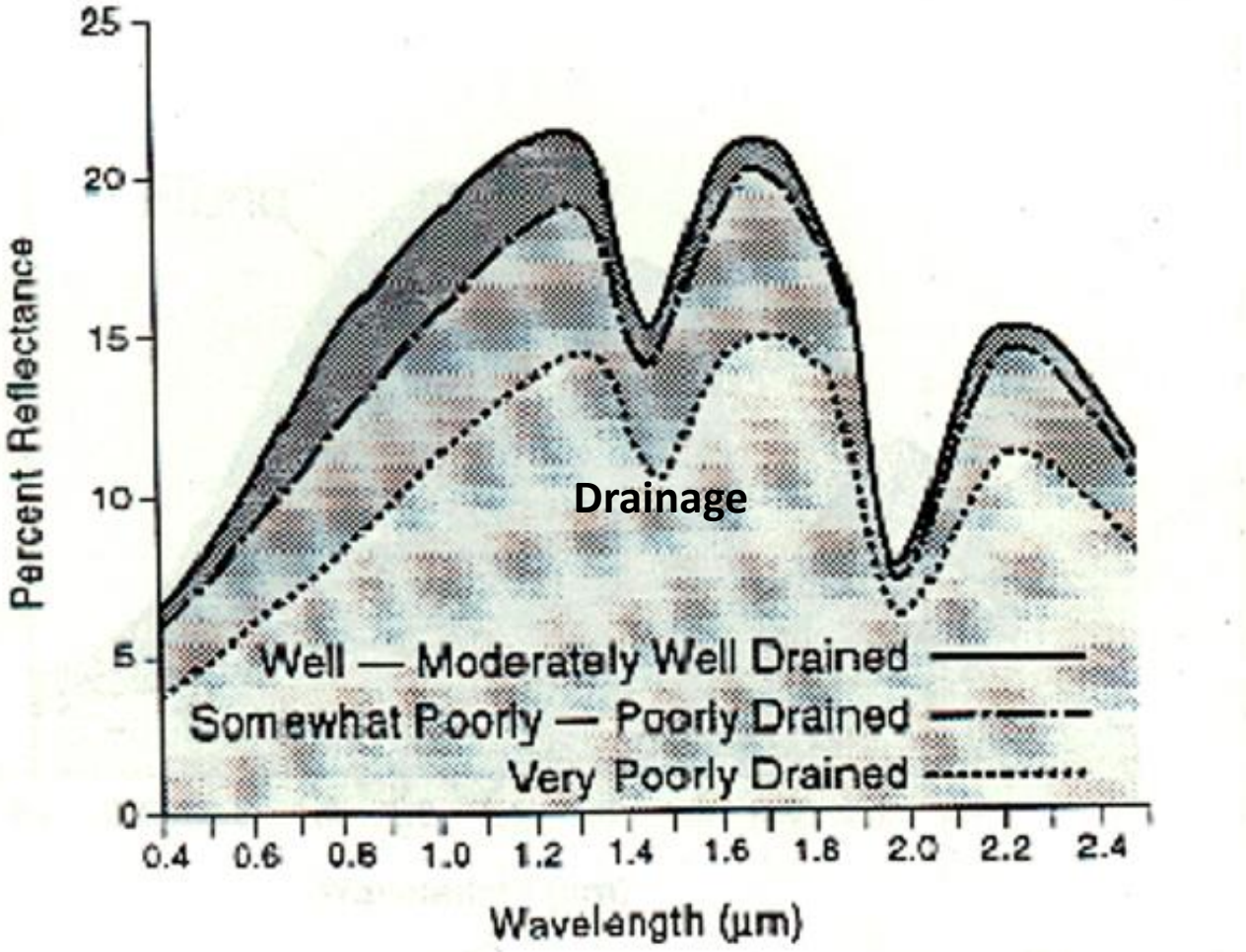


Fig 6: Variation in the spectral reflectance characteristics of soil according to particle size.

http://www.cps-amu.org/sf/notes/m1r-1-8.htm

Soil Drainage



Drainage: removal of surface and sub-surface water from an area

Soil drainage: The speed and ability of water to move through the soil profile, both vertically and horizontally

Soil Texture

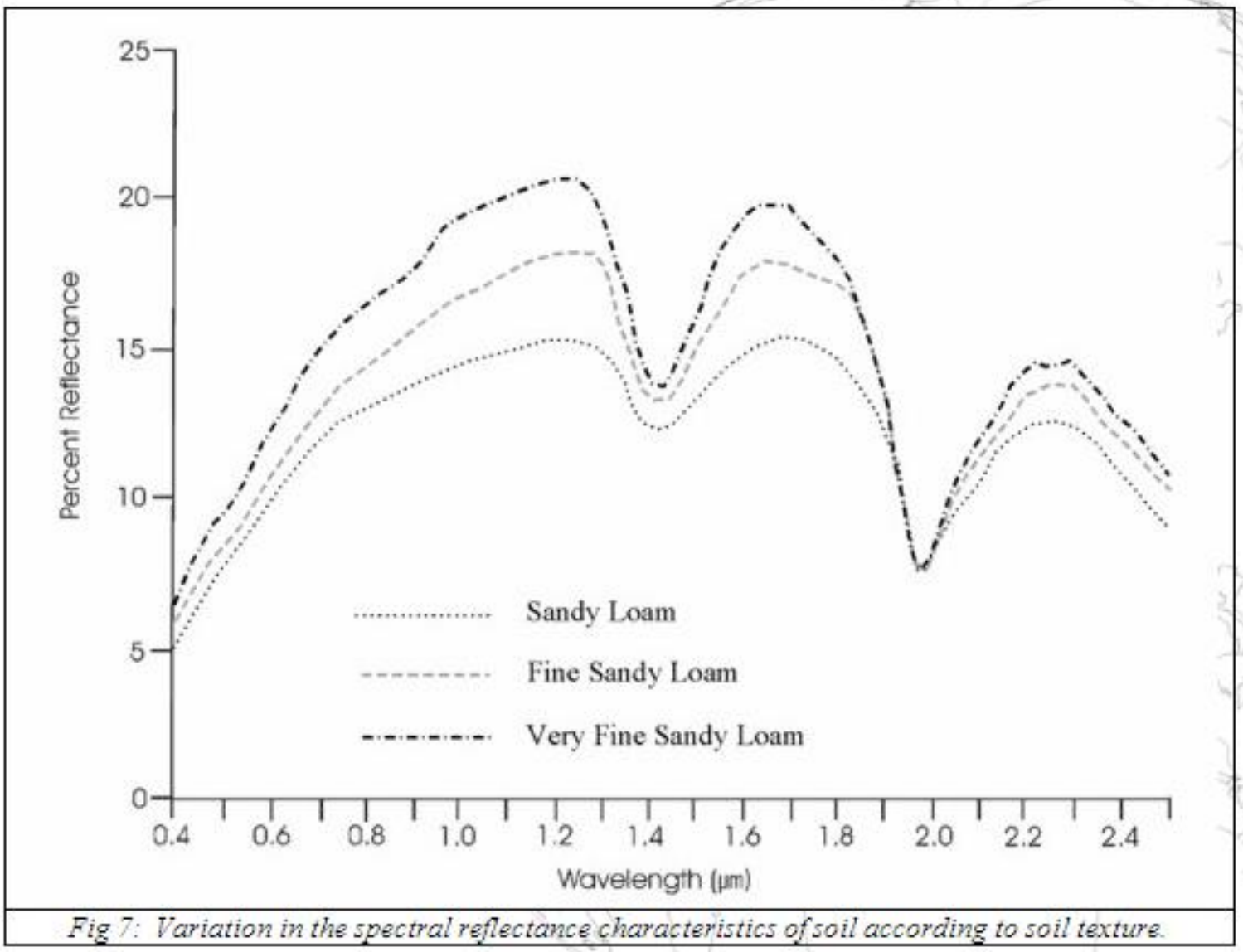


Fig 7: Variation in the spectral reflectance characteristics of soil according to soil texture.

http://www.cps-amu.org/sf/notes/m1r-1-8.htm

Surface Reflectance –Sandy Areas



